

Program : Diploma in Chemical Engineering	
Course Code : 5077	Course Title: Heat Transfer Lab
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3(L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To enable the students to apply heat transfer concepts in practical applications.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Principles of heat transfer		Process heat transfer	4

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Interpret heat transfer by conduction.	9	Applying
CO2	Interpret heat transfer by convection.	12	Applying
CO3	Interpret heat transfer by radiation.	9	Applying
CO4	Interpret heat exchanging equipments.	15	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Interpret heat transfer by conduction.		
M1.01	Determination of thermal conductivity of a metal rod.	3	Applying
M1.02	Determination of thermal conductivity of materials used in composite wall.	3	Applying
M1.03	Open ended experiments	3	Applying
CO2	Interpret heat transfer by convection.		
M2.01	Determination of heat transfer coefficient through natural convection.	3	Applying
M2.02	Determination of heat transfer coefficient through forced convection.	3	Applying
M2.03	Determination of thickness of insulation.	3	Applying
M2.04	Open ended experiments	3	Applying
	Series Test – I		
CO3	Interpret heat transfer by radiation.		
M3.01	Determination of radiation constant of iron cylinder.	3	Applying
M3.02	Determination of radiation constant of iron sphere.	3	Applying
M3.03	Determination of Stefan Boltzman constant.	3	Applying
CO4	Interpret heat exchanging equipments.		
M 4.01	Demonstrate the flow patterns in a shell and tube heat exchanger.	3	Applying

M4.02	Determination of overall heat transfer coefficient of shell and tube heat exchanger.	3	Applying
M4.03	Demonstrate the flow patterns in a double pipe heat exchanger.	3	Applying
M4.04	Determination of overall heat transfer coefficient of double pipe heat exchanger.	3	Applying
M4.05	Open ended experiments	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Heat Transfer; K A Ghavane
R1	Heat transfer and its application; Binay K Dutta
R2	Introduction to Chemical Engineering; Water.L.Badger
R3	Unit operation of Chemical Engineering; Warren.L.McCabe andSmith
R4	Heat transfer; D.D.Tiwari

Online Resources

Sl.No	Website Link
1	https://www.vlab.co.in/broad-area-chemical-engineering